FOR DISCUSSION

Proposal: handwritten math + short answers

12
most relevant papers – to read closely

23
further papers – to skim

~35
HF models checked

~30
datasets checked

Task	Public data?	Best sources	Plan
General math	**Yes**	FERMAT, CHECK-MAT, DrawEduMath, GCSE subset	**Main task (real data)**
Fill-the-gaps	**No**	Only on request (Grabowski 2026); graded typed sets: SciEntsBank, SAF-de	Second task: synthetic handwriting
Free essays	**No**	VisEssay no public download; BRESSAY no grades; Turnitin / ETH: none	Drop (or ask VisEssay authors)

Idea: STACK for handwriting – a fine-tuned open VLM reads the answer, a symbolic / answer-key check grades it

FOR DISCUSSION

What the research says

1 Reading is the bottleneck

39% → 51% → 66%

GPT-4o finds the error from handwriting → printed → text (FERMAT)

87%

of the best grader's errors are misreadings (Levine, AIED 2026)

Same at ETH: handwriting → text is the hardest step

2 Two traps for a STACK-style pipeline

Over-correction: VLMs silently fix student mistakes when transcribing, so the check sees a correct answer

Seong et al. 2026

Reference bias: with the expected answer in the prompt, wrong answers accepted **1 → 193**

Grabowski 2026 · CHECK-MAT

3 Almost nobody fine-tunes

Most papers prompt closed APIs (Gemini, GPT) zero-shot

VEHME (Qwen2.5-VL-7B, LoRA + RL): **31.9%** vs Gemini **64.2%** on FERMAT

Tonmoy 2026 (LoRA): QWK **0.60 → 0.73**

ETH: human-level only if ~70% go back to humans

→ Gap for an open, fine-tuned grader

FOR DISCUSSION

Resources: data, models, key papers

PUBLIC DATASETS

ai4bharat/FERMAT
handwritten math + error labels

CHECK-MAT
122 graded exam solutions (eval)

DrawEduMath
2,030 K-12 student images

MathWriting · HME100K
train the reading stage

SciEntsBank · Beetle · SAF-de
graded short answers → synthetic handwriting

MODELS (HUGGING FACE)

Qwen3-VL-4B / 8B
fine-tuning base, Apache-2.0

DeepSeek-OCR-2 · chandra-ocr-2
OCR stage / baselines

Uni-MuMER-Qwen3-VL-2B · TexTeller
handwritten math → LaTeX

olmOCR-2 · PaddleOCR-VL
more OCR baselines

Mistral OCR
API only – closed baseline, cannot fine-tune

KEY PAPERS

FERMAT (Nath 2025)
benchmark, handwritten math

VEHME (EMNLP 2025)
fine-tuned VLM grader

Levine et al. (AIED 2026)
real exams, error analysis

Henkel et al. 2025
reading vs grading

Seong et al. 2026
VLMs ‘fix’ students

Kortemeyer (ETH) 2024
handwritten math short answers

FOR DISCUSSION

Approach and research questions

Scan
photo of the answer

→

Read
fine-tuned open VLM (LoRA)

→

Check
SymPy / answer key, like STACK

→

Grade
score + confidence

→

Human
only if unsure

1 Fine-tuned small model

Does LoRA on Qwen3-VL-4B close the gap to closed models in error detection on FERMAT?

H1: fine-tuned 4B ≥ zero-shot 8B

2 Two-stage vs end-to-end

Read first, then check (STACK-style) – or grade directly from the image?

H2: end-to-end finds more errors (no 'fixing'); two-stage better on final answers

3 Know when to ask a human

Can the grader abstain on low-confidence answers and stay accurate on the rest?

H3: deferring 20% raises accuracy by ≥ 5 pp

FOR DISCUSSION

Next steps and questions for you

SEMINAR PAPER – CANDIDATES

1. **VEHME (EMNLP 2025)**
   fine-tuning method; same model makes data, reward and evaluation
2. **Henkel et al. 2025**
   reading vs grading; tables contradict the text
3. **Tonmoy et al. 2026**
   handwritten exams + LoRA; private data

NEXT STEPS

- Send paper list; prepare the seminar paper talk + report
- Zero-shot baselines on FERMAT (Qwen3-VL, DeepSeek-OCR-2)
- Prototype synthetic handwriting for short answers
- Fix research question + hypothesis

QUESTIONS FOR YOU

- **Tasks:** math + fill-the-gaps with synthetic handwriting – OK?
- **Essays:** drop, or ask the VisEssay authors for data?
- **Seminar paper:** VEHME or Henkel et al.?

FOR DISCUSSION

Papers (35)

1. FERMAT – Nath 2025 2501.07244
2. VEHME – EMNLP 2025 2510.22798
3. Levine – AIED 2026 2605.19043
4. Yu – college math 2026 2603.00895
5. CHECK-MAT 2025 2507.22958
6. Henkel – Big Picture 2025 2510.05538
7. MathWriting – Google 2024 2404.10690
8. Grabowski – answer tables 2026 2606.11477
9. Fox – GCSE double-marked 2026 2606.24973
10. Kubesch – Austrian essays 2026 2603.06066
11. EssayJudge – ACL Findings 2025 ACL 2025
12. Tonmoy – handwritten exams 2026 2609.14284
13. Caraeni – GPT-4 math exams 2024 2411.05231
14. ETH – handwritten math 2024 2408.11728
15. ETH – thermodynamics 2024 2406.17859
16. ETH – chemistry 2025 2509.10591
17. ETH – IRT confidence 2024 2410.19409
18. ETH – calculus GPT-5 2025 2510.05162
19. ETH – embeddings 2026 2606.28872
20. ETH – GPT-4 on ASAG 2023 2309.09338
21. Perš – engineering exams 2026 2601.00730
22. Vanhoyweghen – human check 2026 2603.13083
23. Pathak – physics olympiad 2026 2608.20521
24. McGinness – bounding boxes 2026 2607.18767
25. Pensieve Grader 2025 2507.01431
26. Sil – answer sheets 2024 2408.12870
27. DrawEduMath – NAACL 2025 2501.14877
28. Aftermath of DrawEduMath 2026 2603.00925
29. Seong – VLMs ‘fix’ students 2026 2604.22774
30. Uni-MuMER – NeurIPS 2025 2505.23566
31. ASAG2024 benchmark 2409.18596
32. RiceChem – AIED 2024 2404.14316
33. Rubric-conditioned grading 2025 2601.08843
34. Fine-tuning for ASAG 2025 2508.04063
35. Essay-BR 2021 2105.09081